

Total No. of Questions : 10]

SEAT No. :

P1870

[4859]-1058

[Total No. of Pages : 2

B.E. (Information Technology)
INFORMATION AND CYBER SECURITY
(2012 Course) (Semester - I) (New)

Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q3 or Q.4,Q.5 or Q.6, Q.7 or Q.8,Q9 or Q.10
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) State the Chinese Remainder theorem with example. [6]
b) In a public key cryptosystem using RSA, given $N=187$ and the encryption key (E) as 17, find out the corresponding private key (D). [4]

OR

- Q2)** a) Draw AES block diagram and explain the steps in detail. [6]
b) Define following. [4]
i) Discrete logarithm
ii) Fermat theorem

- Q3)** a) Explain X.509 standard for Digital Certificate. [6]
b) Explain permutation and substitution steps in DES algorithm. [4]

OR

- Q4)** a) Using Euclidean algorithm calculate [4]
i) GCD (48, 30)
ii) GCD (105, 80)
b) What problem was Kerberos designed to address. Describe Kerberos realm. [6]

P.T.O.

- Q5)** a) Define IKE protocol and illustrate IKE format in detail. [8]
b) Discuss SSL with respect to four phases [8]
i) Establish security capabilities
ii) Server authentication and key exchange
iii) Client authentication and key exchange
iv) Finish

OR

- Q6)** a) Explain various categories of Intrusion Detection system (IDS) [8]
b) How AH and ESP are differs while working under transport and tunnel mode. [8]

- Q7)** a) Describe the classification of Cyber Crime. [10]
b) Define cyber security and information security with example. [6]

OR

- Q8)** a) Explain with example how social engineering is playing wide role in cyber crime. [10]
b) Write a short note on Indian legal perspective. [6]

- Q9)** a) What is SQL injection? Explain in detail. [8]
b) Write short note on: [10]
i) Indian IT act
ii) Different ways of password cracking

OR

- Q10)** Define and differentiate
a) Proxy server and an anonymizer. [6]
b) DOS and DDOS [6]
c) Virus and worm. [6]

